

HealthBridge Wearables Summary

Key Conclusions

Wear OS vs HarmonyOS

- HealthBridge should keep a platform-independent core (Firebase, alerts, caregiver UI).
- Implement separate adapters for Wear OS and Huawei Health Kit rather than rewriting the application.

Huawei Health Kit

- Huawei Health Kit provides APIs for third-party applications with user permission.
- Public documentation includes a Blood Pressure data type (systolic, diastolic, pulse).
- Developers must register with Huawei, enable Health Kit, request the required health scopes, and obtain user consent.

Huawei Watch D2

- Main advantage: inflatable cuff for blood pressure measurement.
- HarmonyOS (not Wear OS).
- Excellent candidate for future HealthBridge support if BP data is accessible through Health Kit on target devices.

Samsung Wear OS Watches

- Best development platform for HealthBridge today.
- Easy Android Studio/Kotlin integration.
- Recommended models: Galaxy Watch Ultra – best battery (3–4 days typical). Galaxy Watch7 – best value. Galaxy Watch4 (refurbished) – lowest-cost development device. **Suggested**

Architecture

HealthBridge Core

- Wear OS Adapter
- Huawei Health Kit Adapter
- Samsung Health Adapter
- Apple HealthKit Adapter
- Shared Firebase backend and caregiver application.

Recommendation

Develop HealthBridge first on Wear OS because of its mature tooling. Then add Huawei support through a dedicated adapter, allowing future access to Huawei health sensors, including blood pressure where supported and authorized.